

W10L1

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Countable Sets.

1. A finite set is countable
2. An infinite set is countable if it has the same no. of elements as that of set of natural numbers.

$$\mathbb{N} = \{0, 1, 2, \dots\}$$

$$A \subseteq \mathbb{N}, A = \{0, 2, 4, 6, \dots\}$$

$$f: \mathbb{N} \rightarrow \underline{A}, \quad f(n) = 2n$$

$$f(0) = \underline{0}, \quad f(1) = \underline{2}, \quad f(2) = \underline{4}, \quad f(3) = \underline{6}, \dots$$

$$f^{-1}(0) = 0, \quad f^{-1}(2) = 1, \quad f^{-1}(4) = 2, \dots$$

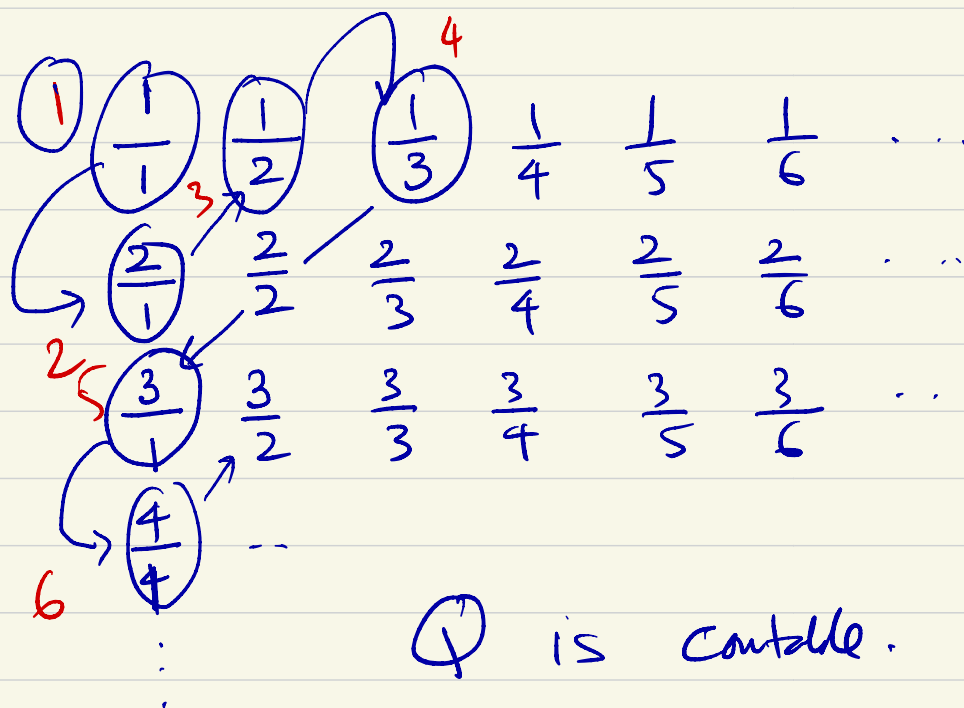
$\mathbb{Q}$ = set of rational number.

$$\mathbb{Q}^+ = \{p/q \mid q \neq 0, \gcd(p, q) = 1, p, q \in \mathbb{N}\}$$

is $\mathbb{Q}$ countable.?

Theorem: $\mathbb{Q}$ is countable.

Diagonalization Argument.



$\mathbb{Q}$ is countable. $\square$

$\mathbb{R}$ = set of real numbers

is $\mathbb{R}$ countable?

Claim: $\mathbb{R}$ is uncountable.

Proof: Assume $\mathbb{R}$ is countable
proof by contradiction

Imagine $I = (0, 1)$

$r_0, r_1, r_2, r_3, \dots$

$$r_0 = 0.\underline{1}372095\dots$$

$$r_1 = 0.52\underline{3}69173\dots$$

$$r_2 = 0.1120201545$$

$$r_3 = 0.752\underline{5}6481\dots$$

$$x = 0.4334$$

" $\sqrt{2}$ is irrational"

Decidable & Recognizable

Theorem: Some languages are not Turing-recognizable.

Outline of the proof

Σ , " Σ^* is countable"

"Set of all languages over Σ is uncountable."

An undecidable Language.

$$A_{TM} = \{ \langle M, w \rangle \mid M \text{ is a TM that accepts } w \}$$

Theorem: " A_{TM} is undecidable."

Proof: Proof by contradiction.

Assume A_{TM} is decidable

Assume there exist a TM H s.t.

H decides A_{TM} .

$$H(\langle M, w \rangle) = \begin{cases} \text{accept} & \text{if } M \text{ accepts } w \\ \text{reject} & \text{if } M \text{ rejects } w \end{cases}$$

Let us create another TM D .

$D =$ "On input $\langle M \rangle$, where M is a TM:

1. Run H on input $\langle M, \langle M \rangle \rangle$

2. Output the opposite of H 's output."

$$D(\langle M \rangle) = \begin{cases} \text{accept} & \text{if } M \text{ does not} \\ & \text{accept } \langle M \rangle \\ \text{reject} & \text{if } M \text{ accepts } \langle M \rangle \end{cases}$$

$$D(\langle D \rangle) = \begin{cases} \text{accept} & \text{if } D \text{ does not} \\ & \text{accept } \langle D \rangle \\ \text{reject} & \text{if } D \text{ accepts } \langle D \rangle \end{cases}$$